CASE STUDY 5: Traffic Fine Calculator

Python Activity-Based Learning

1. Problem Statement

Write a Python program to calculate a traffic fine based on the type of violation: helmet violation ■500, seat-belt violation ■500, and signal violation ■1000.

2. Objectives

• Read the violation type.

• Select the matching fine amount.

• Display the fine.

3. Assumptions

• The user enters one of the listed violation types.

• Fine amounts are fixed for this exercise.

4. Algorithm

1. Start.

2. Read the violation type.

3. If it is helmet, set fine = 500.

4. Else if it is seat belt, set fine = 500.

5. Else if it is signal, set fine = 1000.

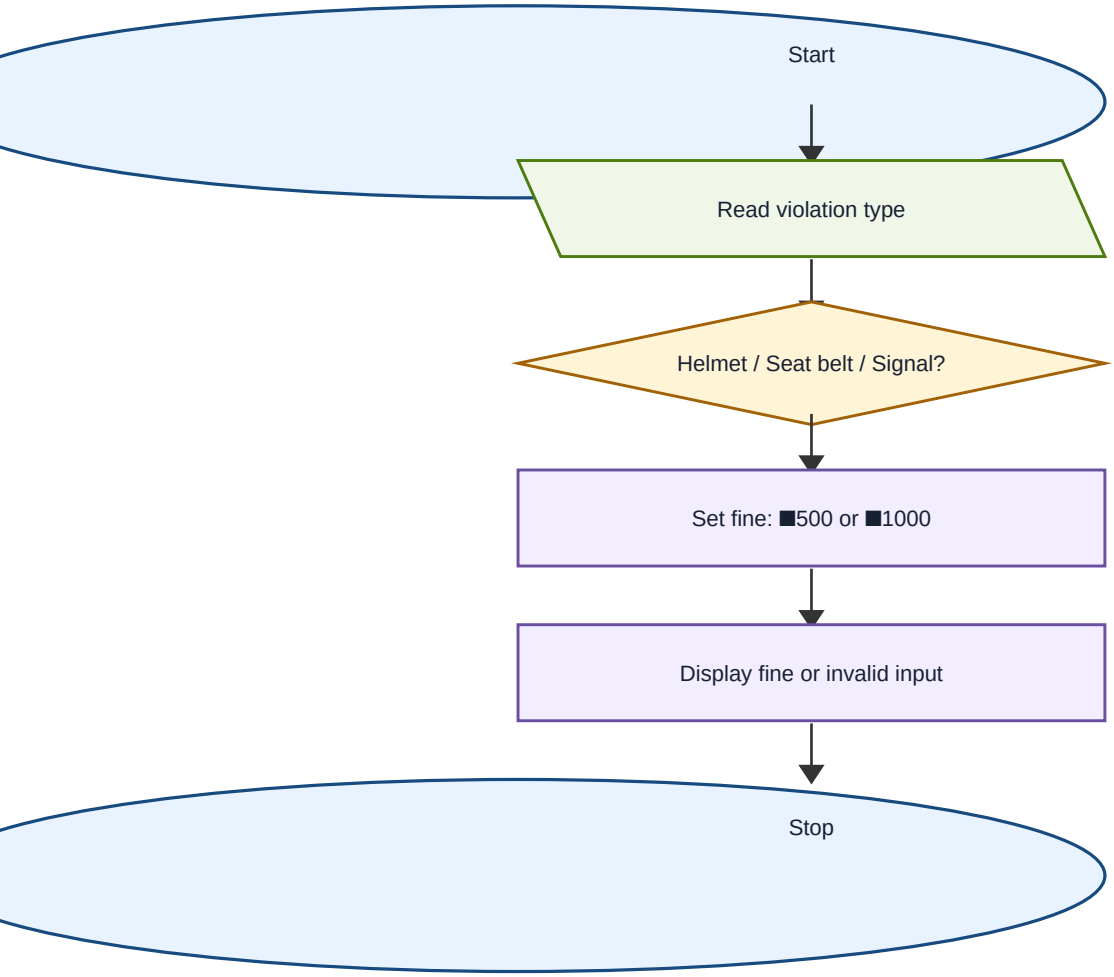
6. Otherwise display Invalid violation.

7. Display the fine when valid.

8. Stop.

5. Flowchart

Oval = Start/Stop; parallelogram = Input; rectangle = Process; diamond = Decision.



6. Pseudocode

```
BEGIN
  INPUT violation
  IF violation == "helmet" THEN
      fine = 500
  ELSE IF violation == "seat belt" THEN
      fine = 500
  ELSE IF violation == "signal" THEN
      fine = 1000
  ELSE
      PRINT "Invalid violation"
  END IF
  IF violation is valid THEN PRINT fine
END
```

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7. Python Source Code

```
violation = input("Enter violation (helmet/seat belt/signal): ").lower()

if violation == "helmet":
    fine = 500
    print("Fine: ■", fine)
elif violation == "seat belt":
    fine = 500
    print("Fine: ■", fine)
elif violation == "signal":
    fine = 1000
    print("Fine: ■", fine)
else:
    print("Invalid violation type")
```

8. Test Cases

Input	Expected Output
helmet	Fine: ■500
signal	Fine: ■1000
parking	Invalid violation type

9. Sample Output

Run the Python program and enter the values from one of the test cases above. The displayed result should match the expected output.

10. Conclusion

The program uses if-elif-else statements to select and display a fine based on the violation entered.